Supporting Information for Non-Redundant and Symmetry-Adapted Coordinates for Black-Box Molecular Structure Refinement

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Glycine II Functional-Constraint Benchmark

Table 1 reports the Gly-II DPCS3-constrained benchmark discussed in the main text. The fit uses moments of inertia as observables, but the diagnostic table is reported in MHz. The corrected experimental constants are $B_e = B_0 + \Delta B_{\mathrm{vib}}$ with the Gly-IIn zero-point corrections.

Table 1: Complete rotational-constant comparison for glycine II. Corrected experimental constants, calculated constants, and differences are in MHz.

Isotopologue	Constant	Corrected exp.	Calculated	Difference
parent	A	10195.5900	10196.2607	-0.6707
parent	B	4105.3670	4105.3251	0.0419
parent	C	3039.8450	3039.8159	0.0291
N15	A	10194.1150	10194.6721	-0.5571
N15	B	3994.8380	3994.8046	0.0334
N15	C	2978.7993	2978.7971	0.0022
C13_methylene	A	10060.9700	10061.6298	-0.6598
C13_methylene	B	4088.6720	4088.5073	0.1647
C13_methylene	C	3018.8730	3018.8495	0.0235
C13_carboxyl	A	10195.6980	10195.0945	0.6035
C13_carboxyl	B	4093.0000	4093.1241	-0.1241
C13_carboxyl	C	3033.1070	3033.1532	-0.0462
O18_carbonyl	A	10064.2400	10063.7580	0.4820
O18_carbonyl	B	3929.9160	3929.9078	0.0082
O18_carbonyl	C	2931.7130	2931.8037	-0.0907
O18_hydroxyl	A	9580.9720	9580.9886	-0.0166
O18_hydroxyl	B	4090.8498	4090.8965	-0.0467
O18_hydroxyl	C	2975.1370	2975.1377	-0.0007
OD_NH2	A	9792.2000	9792.2302	-0.0302
OD_NH2	B	4096.1490	4096.1104	0.0386

Isotopologue	Constant	Corrected exp.	Calculated	Difference
OD_NH2	C	2998.1238	2998.0987	0.0251
CD2	A	9145.3030	9145.3053	-0.0023
CD2	B	4020.3863	4020.3836	0.0027
CD2	C	2947.9923	2947.9913	0.0010
OH_ND2	A	9927.6990	9927.6852	0.0138
OH_ND2	B	3716.3758	3716.3613	0.0145
OH_ND2	C	2841.7341	2841.7204	0.0137
OD_ND2	A	9542.6600	9542.6581	0.0019
OD_ND2	B	3710.5050	3710.5504	-0.0454
OD_ND2	C	2806.0910	2806.1569	-0.0659

Cyclopentadiene Benchmark

Table 2 reports the complete comparison between corrected experimental rotational constants and constants calculated from the final semiexperimental structure. The workflow uses generalized internal coordinates (GICs) or symmetry-adapted Cartesian coordinates generated within `SEfit`. The data set contains 14 isotopologues and all three principal rotational constants for each isotopologue. The refinement uses the same spectroscopic information derived from the `MSR` molecular-structure-refinement workflow, but the molecular structure is represented only by Cartesian coordinates and automatically generated non-redundant symmetry-adapted GICs; no Z-matrix or manually constructed internal-coordinate path is used.

Table 2: Complete rotational-constant comparison for the cyclopentadiene benchmark derived from MSR data. Corrected experimental and calculated constants are in MHz.

Isotopologue	Component	Corrected exp.	Calculated	Difference
msr_symm_01	A	8489.154993	8489.232117	-0.077124
msr_symm_01	B	8292.100993	8292.093033	0.007960
msr_symm_01	C	4305.664997	4305.658417	0.006580
msr_symm_02	A	8292.098993	8292.093033	0.005960
msr_symm_02	B	8280.278992	8280.319224	-0.040231
msr_symm_02	C	4251.270992	4251.257436	0.013556
msr_symm_03	A	8482.790995	8482.866229	-0.075234
msr_symm_03	B	8105.052002	8105.082639	-0.030637
msr_symm_03	C	4253.082997	4253.084517	-0.001520
msr_symm_04	A	8408.381002	8408.358013	0.022989
msr_symm_04	B	8172.557990	8172.606506	-0.048516
msr_symm_04	C	4252.643011	4252.628488	0.014523
msr_symm_05	A	8408.381002	8408.358024	0.022978

Isotopologue	Component	Corrected exp.	Calculated	Difference
msr_symm_05	B	8172.557990	8172.606496	-0.048506
msr_symm_05	C	4252.643011	4252.628488	0.014523
msr_symm_06	A	8195.611995	8195.593227	0.018768
msr_symm_06	B	7918.397999	7918.289812	0.108187
msr_symm_06	C	4178.352982	4178.398406	-0.045424
msr_symm_07	A	8195.611995	8195.612584	-0.000589
msr_symm_07	B	7918.397999	7918.308530	0.089469
msr_symm_07	C	4178.352982	4178.398587	-0.045604
msr_symm_08	A	8477.175001	8477.228373	-0.053372
msr_symm_08	B	7650.516996	7650.483876	0.033120
msr_symm_08	C	4123.171991	4123.147057	0.024934
msr_symm_09	A	8477.175001	8477.228367	-0.053366
msr_symm_09	B	7650.516996	7650.483883	0.033113
msr_symm_09	C	4123.171991	4123.147057	0.024933
msr_symm_10	A	8371.829003	8371.953330	-0.124328
msr_symm_10	B	7735.107991	7735.111002	-0.003010
msr_symm_10	C	4122.272002	4122.241235	0.030767
msr_symm_11	A	8371.829003	8371.953344	-0.124341
msr_symm_11	B	7735.107991	7735.110990	-0.002998
msr_symm_11	C	4122.272002	4122.241235	0.030767
msr_symm_12	A	7057.547996	7057.656561	-0.108565
msr_symm_12	B	6730.362994	6730.334074	0.028920
msr_symm_12	C	3555.255987	3555.260397	-0.004410
msr_symm_13	A	7057.547996	7057.672705	-0.124709
msr_symm_13	B	6730.362994	6730.349532	0.013462
msr_symm_13	C	3555.255987	3555.260180	-0.004193
msr_symm_14	A	6656.949995	6656.991310	-0.041315
msr_symm_14	B	6653.764996	6653.536122	0.228874
msr_symm_14	C	3469.286997	3469.303412	-0.016415

Nitrobenzene Planar Benchmark

Nitrobenzene is a planar aromatic benchmark. The least-squares fit used the independent AB moment pair, corresponding to (I_a, I_b), while the C constants were recomputed from the optimized Cartesian geometry and retained as diagnostic quantities. Table 3 reports the complete diagnostic comparison.

Table 3: Complete rotational-constant comparison for the nitrobenzene planar benchmark. Corrected experimental and calculated constants are in MHz. The fit used the independent AB pair; the C component is diagnostic.

Isotopologue	Component	Corrected exp.	Calculated	Difference
parent	A	3995.209000	3995.211821	-0.002821
parent	B	1296.305300	1296.307106	-0.001806
parent	C	979.000500	978.740045	+0.260455
N15	A	3995.193000	3995.211821	-0.018821
N15	B	1287.445000	1287.445036	-0.000036
N15	C	973.938200	973.679678	+0.258522
O18	A	3925.992000	3925.984021	+0.007979
O18	B	1264.689700	1264.689116	+0.000584
O18	C	956.803400	956.552095	+0.251305
6C13	A	3995.172000	3995.211821	-0.039821
6C13	B	1274.812600	1274.812638	-0.000038
6C13	C	966.688800	966.436980	+0.251820
6D	A	3995.241000	3995.211821	+0.029179
6D	B	1253.544800	1253.544803	-0.000003
6D	C	954.416100	954.164457	+0.251643
2C13	A	3949.267000	3949.255685	+0.011315
2C13	B	1295.514600	1295.514106	+0.000494
2C13	C	975.768500	975.508297	+0.260203
2D	A	3856.605000	3856.604007	+0.000993
2D	B	1296.294600	1296.294437	+0.000163
2D	C	970.447500	970.190734	+0.256766
3C13	A	3950.485000	3950.473551	+0.011449
3C13	B	1284.693900	1284.693395	+0.000505
3C13	C	969.688400	969.433703	+0.254697
3D	A	3858.465000	3858.463986	+0.001014
3D	B	1276.881200	1276.881052	+0.000148
3D	C	959.645200	959.390171	+0.255029

The Kraitchman diagnostic for nitrobenzene is reported in Table 4. The comparison uses absolute substitution coordinates because the Kraitchman equations do not determine coordinate signs. The diagnostic was not used in the least-squares objective.

Table 4: Kraitchman diagnostic for the nitrobenzene planar benchmark. Absolute Kraitchman substitution coordinates and absolute fitted coordinates are in Å.

Isotopologue	Atom	Axis	Kraitchman	Fit	Difference
N15	2 N	a	1.64701	1.64723	-0.00022
N15	2 N	b	0.01795	0.00000	+0.01795
N15	2 N	c	0.01379	0.00000	+0.01379
O18	13 O	a	2.22302	2.21369	+0.00933
O18	13 O	b	1.06338	1.08300	-0.01962
O18	13 O	c	0.00896	0.00000	+0.00896
6C13	11 C	a	2.56993	2.56999	-0.00005
6C13	11 C	b	0.03786	0.00000	+0.03786
6C13	11 C	c	0.00000	0.00000	+0.00000
6D	12 H	a	3.65015	3.65027	-0.00012
6D	12 H	b	0.00000	0.00000	+0.00000
6D	12 H	c	0.01902	0.00000	+0.01902
2C13	3 C	a	0.48915	0.48830	+0.00085
2C13	3 C	b	1.21604	1.21671	-0.00066
2C13	3 C	c	0.00000	0.00000	+0.00000
2D	4 H	a	0.05793	0.06125	-0.00332
2D	4 H	b	2.13423	2.13424	-0.00001
2D	4 H	c	0.00000	0.00000	+0.00000
3C13	7 C	a	1.88184	1.87669	+0.00515
3C13	7 C	b	1.19989	1.20775	-0.00786
3C13	7 C	c	0.00000	0.00000	+0.00000
3D	8 H	a	2.43739	2.41687	+0.02052
3D	8 H	b	2.11909	2.14308	-0.02399
3D	8 H	c	0.03090	0.00000	+0.03090

Azulene Comparison with the Reference Refinement

The azulene calculation used the MSR-comparable isotopologue subset and the same fixed structural information as primitive-coordinate constraints. The fixed C–H information in the MSR fit corresponds to C1–H11, its C_{2v} partner C3–H13, and C2–H12; the remaining C–H distances are refined. In the original MSR Z-matrix, one ring-closing C–C distance is a derived quantity rather than a primary coordinate, whereas the Cartesian/GIC workflow evaluates all reported topological distances from the final Cartesian geometry. All coordinates were generated automatically from Cartesian structures; no Z-matrix was used. Because azulene is planar, the fit used the independent AB moment pair and retained the C constants as diagnostics. Tables 5 and 6 report the complete rotational-constant comparison and the full structural comparison with the reference MSR values, respectively.

Table 5: Complete rotational-constant comparison for the azulene MSR-comparable subset. Corrected experimental and calculated constants are in MHz.

Isotopologue	Component	Corrected exp.	Calculated	Difference
parent	A	2862.097939	2862.075372	+0.022567
parent	B	1262.517835	1262.559239	-0.041404
parent	C	876.138923	876.087228	+0.051695
azulene-4-d1	A	2752.561850	2752.579320	-0.017470
azulene-4-d1	B	1262.294384	1262.218223	+0.076161
azulene-4-d1	C	865.456501	865.387542	+0.068959
azulene-5-d1	A	2794.488840	2794.483227	+0.005613
azulene-5-d1	B	1241.273362	1241.275392	-0.002030
azulene-5-d1	C	859.559161	859.497208	+0.061953
azulene-6-d1	A	2862.003650	2862.075372	-0.071722
azulene-6-d1	B	1223.400793	1223.394588	+0.006205
azulene-6-d1	C	857.105233	857.048897	+0.056336
azulene-4,8-d2	A	2649.532362	2649.523242	+0.009120
azulene-4,8-d2	B	1262.005103	1261.903897	+0.101206
azulene-4,8-d2	C	854.871156	854.788696	+0.082460
azulene-5,7-d2	A	2725.125973	2725.133539	-0.007566
azulene-5,7-d2	B	1221.780703	1221.778074	+0.002629
azulene-5,7-d2	C	843.602156	843.573085	+0.029071
azulene-1-13C	A	2841.331150	2841.459456	-0.128306
azulene-1-13C	B	1251.448996	1251.449765	-0.000769
azulene-1-13C	C	868.849088	868.805922	+0.043166
azulene-2-13C	A	2862.093080	2862.075372	+0.017708
azulene-2-13C	B	1240.233194	1240.296442	-0.063248
azulene-2-13C	C	865.322674	865.309646	+0.013028
azulene-9-13C	A	2853.194180	2852.904191	+0.289989
azulene-9-13C	B	1261.606550	1261.641703	-0.035153
azulene-9-13C	C	874.835186	874.784969	+0.050217
azulene-4-13C	A	2821.753737	2821.770671	-0.016934
azulene-4-13C	B	1261.608919	1261.591491	+0.017428
azulene-4-13C	C	871.875087	871.811446	+0.063641
azulene-5-13C	A	2836.950530	2837.099313	-0.148783
azulene-5-13C	B	1251.104248	1251.070937	+0.033311
azulene-5-13C	C	868.277606	868.215431	+0.062175
azulene-6-13C	A	2862.033150	2862.075372	-0.042222
azulene-6-13C	B	1243.247466	1243.219286	+0.028180
azulene-6-13C	C	866.765213	866.731282	+0.033931

Table 6: Complete structural comparison for azulene. Bond lengths are in Å; angles are in degrees. The MSR column corresponds to the reference values after matching the $R(1,9)/R(4,5)$ pair to the Cartesian/GIC atom labels.

Parameter	Unit	`SEfit`	σ	MSR	Difference
$R(1,9)$	Å	1.39745	0.00599	1.39900	-0.00155
$R(4,10)$	Å	1.37923	0.01201	1.37800	+0.00123
$R(4,5)$	Å	1.40122	0.00653	1.40100	+0.00022
$R(5,6)$	Å	1.39054	0.00442	1.39200	-0.00146
$R(4,14)$	Å	1.08356	0.00220	1.08700	-0.00344
$R(5,15)$	Å	1.08133	0.00230	1.08200	-0.00067
$R(6,16)$	Å	1.08117	0.00441	1.08200	-0.00083
$R(1,11)$	Å	1.08116	0.00000	1.08000	+0.00116
$R(2,12)$	Å	1.08236	0.00000	1.08100	+0.00136
$R(1,2)$	Å	1.39593	0.00584	1.40400	-0.00807
$R(9,10)$	Å	1.50860	0.01549	1.50000	+0.00860
$A(1,9,10)$	degree	106.13001	0.52104	106.50000	-0.36999
$A(5,6,7)$	degree	130.09492	0.17695	129.90000	+0.19492
$A(6,5,15)$	degree	115.90333	0.30686	115.70000	+0.20333
$A(4,5,6)$	degree	128.68339	0.00000	128.68000	+0.00339
$A(5,4,10)$	degree	128.79958	0.00000	128.81000	-0.01042
$A(9,1,11)$	degree	125.13843	0.00000	125.15000	-0.01157
$A(10,4,14)$	degree	115.31629	0.00000	115.31000	+0.00629
$A(1,2,12)$	degree	125.06740	0.00000	125.07000	-0.00260
$A(1,2,3)$	degree	109.86520	0.00000	109.71000	+0.15520
$A(2,1,9)$	degree	108.93739	0.52104	108.66000	+0.27739
$A(4,10,9)$	degree	127.46958	0.08847	127.58000	-0.11042
$A(5,6,16)$	degree	114.95254	0.08847	115.07000	-0.11746

Reproducibility Package

All semiexperimental runs write the same output contract: plain-text reports, HTML reports (HyperText Markup Language), final Cartesian geometry, working-coordinate and primitive-parameter tables, rotational-constant and residual tables, covariance/correlation and influence diagnostics, singular-value-decomposition (SVD) diagnostics of the weighted Jacobian, constraint audit tables, diagnostic warning tables, Hessian eigenvalues, Kraitchman diagnostics, optional leave-one-isotopologue-out diagnostics, LaTeX table fragments, a restartable checkpoint, and a manifest. The `semiexp_manifest.json` file stores input and output hashes, selected coordinate model, rotational components, constraints, parameter classes, robust-loss settings, checkpoint location, and convergence diagnostics.

Table 7: Input files and coordinate choices for the benchmark calculations. Paths are relative to the Merlino repository unless an absolute legacy input path is shown. The same runs can be launched from the graphical user interface (GUI) by selecting the Python backend, the indicated coordinate model, and the same input files.

Case	Input and constraints	Coordinate choice
Glycolaldehyde	`working/semiexp/glycolaldehyde/glycolaldehyde_reference.mse.toml`; no fixed primitive constraints.	GIC, ABC
Glycine II	`benchmarks/semiexp_msr/inputs/glycine/glycine_II_dpcs3_table5_constraints.msr`; non-Cs DPCS3 Cartesian geometry, Gly-IIn vibrational corrections, and five Gaussian-style functional constraints with `Value=` targets.	GIC, ABC
Cyclopentadiene	`benchmarks/semiexp_msr/inputs/cyclopentadiene/parent.xyz` and `benchmarks/semiexp_msr/inputs/cyclopentadiene/isotopologues.toml`; no fixed primitive constraints.	symmetry-Cartesian, ABC
Nitrobenzene	Legacy-compatible MSR input `benchmarks/semiexp_msr/inputs/nitrobenzene/nitrobenzene_all_msr.inp`; frozen variables are converted to primitive constraints and no Z-matrix is used after import; planar components selected automatically.	GIC, auto planar pair
Azulene	`working/semiexp/azulene/azulene_msr_like.mse.toml`; C–H distances and selected valence angles fixed as primitive constraints.	GIC, AB
Norcamphor	Reference-constrained norcamphor input; 30 primitive hydrogen constraints reproduce the reduced-dimensionality MSR information.	GIC, ABC

Canonical commands.

```
python -m merlino semiexp \
  --job working/semiexp/glycolaldehyde/glycolaldehyde_reference.mse.toml \
  --outdir working/semiexp/glycolaldehyde/run_reference \
  --coordinate-model gic
```

```
CYC=benchmarks/semiexp_msr/inputs/cyclopentadiene
OUT=working/semiexp/cyclopentadiene/run_msr_symm_cartesian_symmetry
python -m merlino semiexp \
  --xyz "$CYC/parent.xyz" \
  --observations "$CYC/isotopologues.toml" \
  --outdir "$OUT" \
  --coordinate-model cartesian_symmetry

GLY=benchmarks/semiexp_msr/inputs/glycine
python -m merlino semiexp \
  --job "$GLY/glycine_II_dpcs3_table5_constraints.msr" \
  --outdir working/semiexp/glycine_II_dpcs3_table5_repo_gic_fortran \
  --backend fortran77 \
  --coordinate-model gic \
  --observable moments \
  --rotational-components auto

NIT=benchmarks/semiexp_msr/inputs/nitrobenzene/nitrobenzene_all_msr.inp
python -m merlino semiexp \
  --job "$NIT" \
  --outdir nitrobenzene_merlino_pair_auto \
  --coordinate-model gic \
  --rotational-components auto

python -m merlino semiexp \
  --job working/semiexp/azulene/azulene_msr_like.mse.toml \
  --outdir working/semiexp/azulene/run_msr_like_planar_AB \
  --coordinate-model gic \
  --rotational-components AB

python -m merlino semiexp \
  --job working/semiexp/norcamphor/reference_constraints.mse.toml \
  --outdir working/semiexp/norcamphor/run_reference_constraints \
  --coordinate-model gic

python -m merlino semiexp-benchmark --paper
```